

Plotting Data from a Database Transcript

Let us explore the other features of plotting from the data frames. What did we do? We just connected with the data from the web, imported that into a data frame called `df.weather` and tried to understand how the maximum temperature is varying over Seattle by setting the date as the x-axis. The index function arranges the date as the time series value, which means it is changing over a period of time and then you are plotting the temperature.

Now let us see what other columns are there in the data frame that we have created. So `df.weather` is the data frame that we have imported from the website. And then now we are looking at the... Okay, so I just made a mistake, small mistake.

So be careful about the syntax. It is just `df.weather`. So these are the other variables that you have in the particular data frame. Now the task is let us see how does this perform when you want to look at the precipitation.

So how does precipitation change over the Seattle region? So I am just copying the code which is required for plotting the diagram because all others are not required to be repeated. Just inserting one more code snippet. Instead of temperature max, I am typing precipitation, which indicates the precipitation or rainfall.

So I am saying daily. The title is changed suitably by typing daily precipitation in Seattle. Instead of temperature, I am typing precipitation in mm.

And date. And always remember the colour of the graphic should reflect the mood of the data that you are reflecting. So rainfall generally doesn't associate with red colour.

So we are using blue and just running the graph. So this is how you get. So the point here is you have the time, the same date axis and the precipitation on y-axis and you are waiting to graph it and you have the temperature.

Now the ask is little different. Instead of having two different graphs, I want to see the temperature variation, rainfall variation and maybe the wind variation in the same graph in Seattle. So what we are using now is the subplot function, which I have already seen in the last class.

But we will now try to do that with the shared x-axis. The point here is the x-axis, which is the date range is common for all the phenomenon of the temperature or precipitation or wind that you want to observe. So go back to the data frame.

So you have the date and you have the four or five values that are there in the table. Now you want to present them together. So they have the same common x-axis.

So the subplots share the common x-axis or common y-axis depending upon the use case. Let us say you have a condition where you are trying to see the target achievement, which will be 0 to 100 percent or above 100 percent. Let us say and then you have the different regions or different products that you have within the regions.

So that you want to see in that case, you want to share the y-axis instead of the x-axis. So in this case, we are sharing the x-axis. And how are we going to create this? So instead of, I want to create a 3 by 1 grid, which is three rows and one column.

Always like in matrix algebra, first element always represents the rows, second element always represents the column. m cross n rows into columns, right? So what am I doing? I am sharing the x-axis. Share x means, share x is equal to 2, which is sharing the x-axis, which is quite intuitive.

I am saying a figure and the axis. This is the command, figure is the figure, axis is the axis. So plot dot subplot is the function that will give you this.

So I am just running this part, right? I am just running this selection. So what is, look at what is getting created. You are getting a plot, right? Which figure, which shows the shared x-axis and the shared, I mean, and different y-axis.

The way to read this is, the x-axis is below and all these three graphics will have the same x-axis. So you can compare in this particular section of x-axis, what these three variables are performing as. This is something like sub-multiples features in the Power BI that we have seen earlier.

So I have done the three subplots and now I am plotting for each of the graphic. I am passing, I am entering the command. And mind you, in Python, the indexing starts with zero, not with one.

So the first element is always represented with zero, which would represent the first thing. One goes to the second, two goes to the three. So first we are plotting the temperature and I am using crimson red.

Then I am using precipitation, right? And then I am using the royal blue. So now look at the syntax. There is a slight difference that you have here, right? With the `df.plot.figure` and the way that we have used this particular function, right? So you have `df.weather, temperature, max, plot.colour`. The code or the syntax is a little different here.

You are using `axis.plot`. You are telling, use the x-axis as `df.weather.index`, which we have set to date earlier. So please recall, we have set the index to date. So we are using that particular thing, which is the index.



Then we are using `df.weather` again, subsetting the temperature max column and then colour is crimson. Rest are all titles and labels, but this is the most important thing. You can also call a temperature minimum or something like that.

So you can also do that. Instead of this, I am doing temperature min. So I hope it is, yeah, temperature min.

And I am saying, instead of maximum temperature, I am doing the minimum temperature in this. And I am, yeah, so instead of putting them together and just 0, sorry, 1, 1, 1. So you have the, so this is a mistake I made. I should make it minimum temperature.

So you can see, right, the maximum temperature ranges from 0 to 30 degrees. Minimum range is minus 5 to 15 degrees centigrade. And then you have the rainfall all in single graph, right? And you are able to see that generally when the temperatures are high, you don't have much precipitation happening.

That kind of conclusion you can come to or understand the different seasons in Seattle or the years, how are they changing? What is the fluctuation of the temperature? So all this can be done using the subplot function where you are trying to share the x-axis and plot the different columns in the data frame. Okay. We will see more examples in the live class and keep practising till then from the documentation and the examples that we are loading.

Thank you.

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